

Supplementary Experimental Results for B $\otimes$ X

Due to space constraints in the main manuscript, we only present representative results. This paper serves as supplementary material, providing a comprehensive display of all experimental results presented in the manuscript.

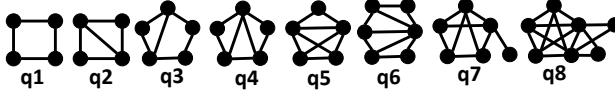


Fig. 1. Global queries.

1 OVERALL PERFORMANCE

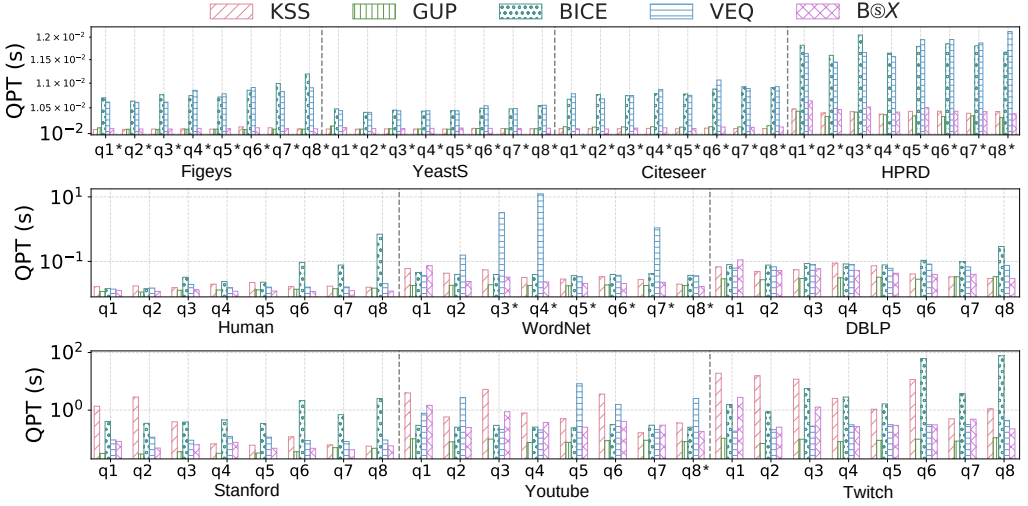


Fig. 2. QPT of different methods with output limit.

Under the QPT metric. To further compare the performance differences between B $\otimes$ X and other methods, we conduct experiments with an output limit of 10^5 for comparisons using the QPT metric. Figure 2 reports the QPT of global queries from Figure 1 across all datasets. Queries with zero embeddings are marked with an asterisk (*) in the figure.

2 SCALABILITY EVALUATION

2.1 Varying the Density of Query Graphs

To investigate the impact of query graph density on algorithm performance, we generate query graphs across varying densities. We use the average degree of a query graph as a measure of its density. Figure 3 displays the performance of all six methods across various densities of query graphs across all dataset, where D_i denotes the range $[i, i+1)$.

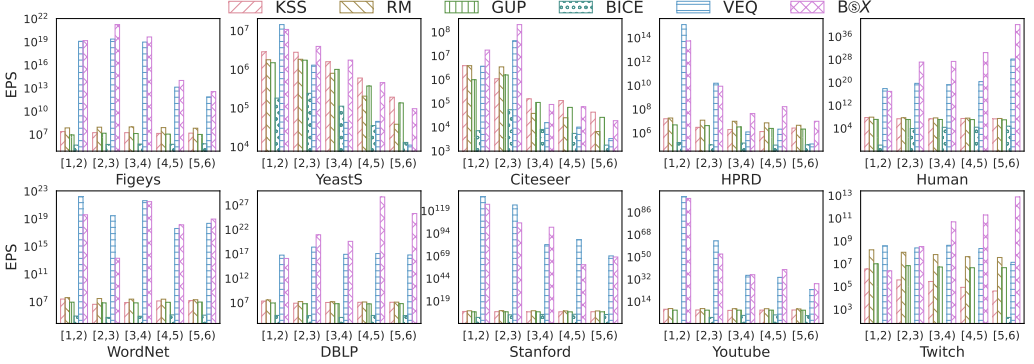


Fig. 3. Performance under different query graph densities.

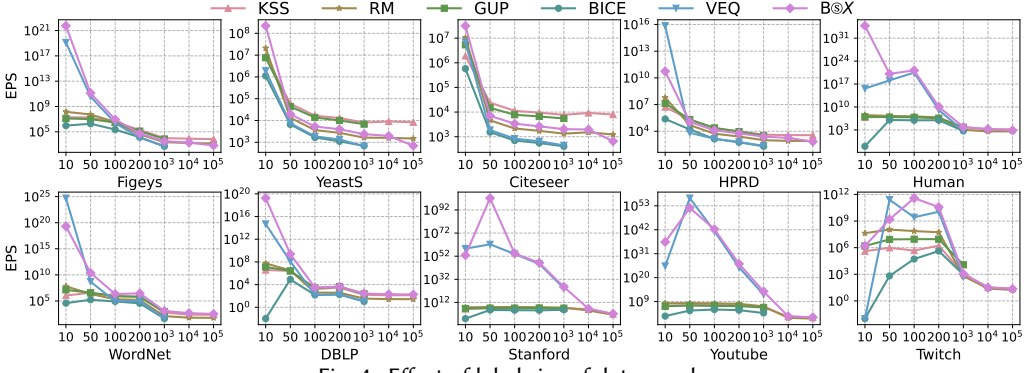


Fig. 4. Effect of label size of data graphs.

2.2 Varying the Label Size of Data Graphs

Figure 4 illustrates the performance across seven label sizes: 10, 50, 100, 200, 10^3 , 10^4 , and 10^5 . Due to memory overflow issues with BICE, VEQ, and GUP for large label size, they are unable to handle cases involving 10^4 and 10^5 labels.

2.3 Varying the Size of Query Graph

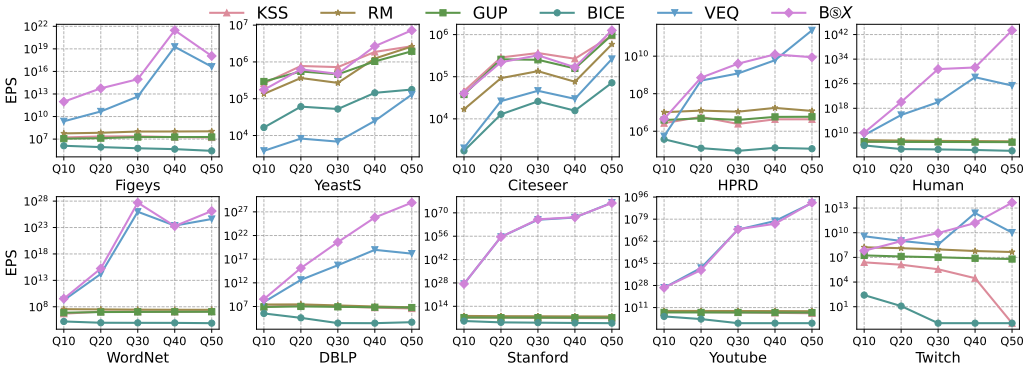


Fig. 5. Effect of query graph size.

We have examined the performance of various algorithms across query graph sizes: 10, 20, 30, 40, and 50. Figure 5 presents the EPS for these query sizes across all datasets.

2.4 Varying the size of Data Graph

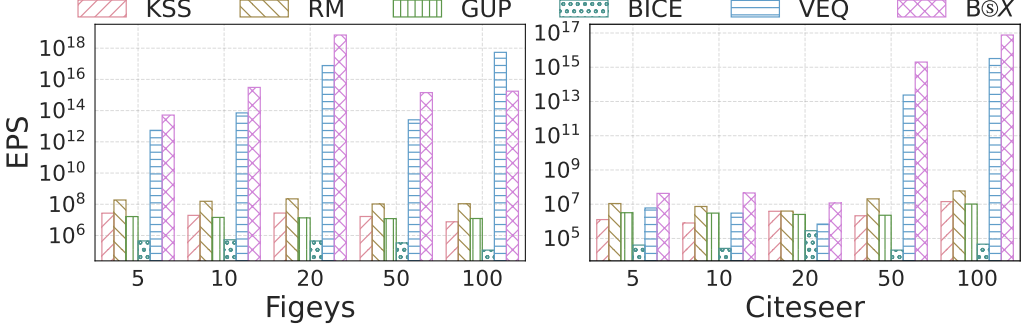


Fig. 6. Effect of data graph size (1).

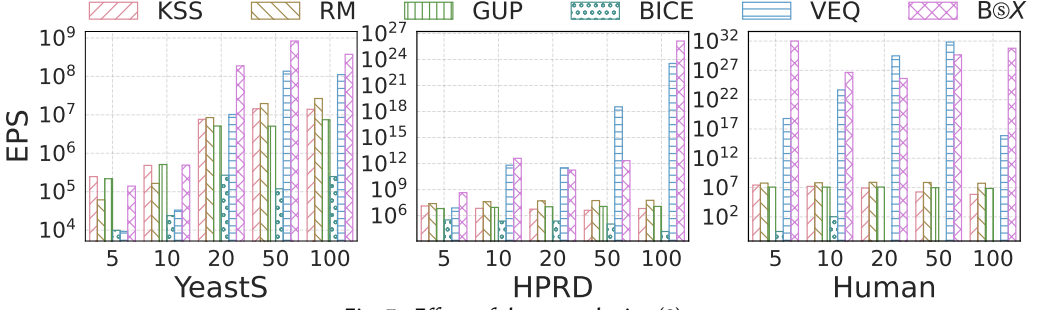


Fig. 7. Effect of data graph size (2).

To evaluate the scalability of B $\otimes$ X, we evaluate the performance of various algorithms across data graphs of different sizes. We generate synthetic graphs by expanding the small-scale graphs (with a total number of vertices less than 10,000) Figeys, YeastS, Citeseer, HPRD, and Human using scaling factors of 5, 10, 20, 50, and 100 with EvoGraph. For each scaled data graph, we generate 500 query graphs. Figure 6, 7 presents the EPS results of all methods on these synthetic data graphs.